

Data Flow Diagram

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. The completed FitBuddy DFD is provided on the next page.

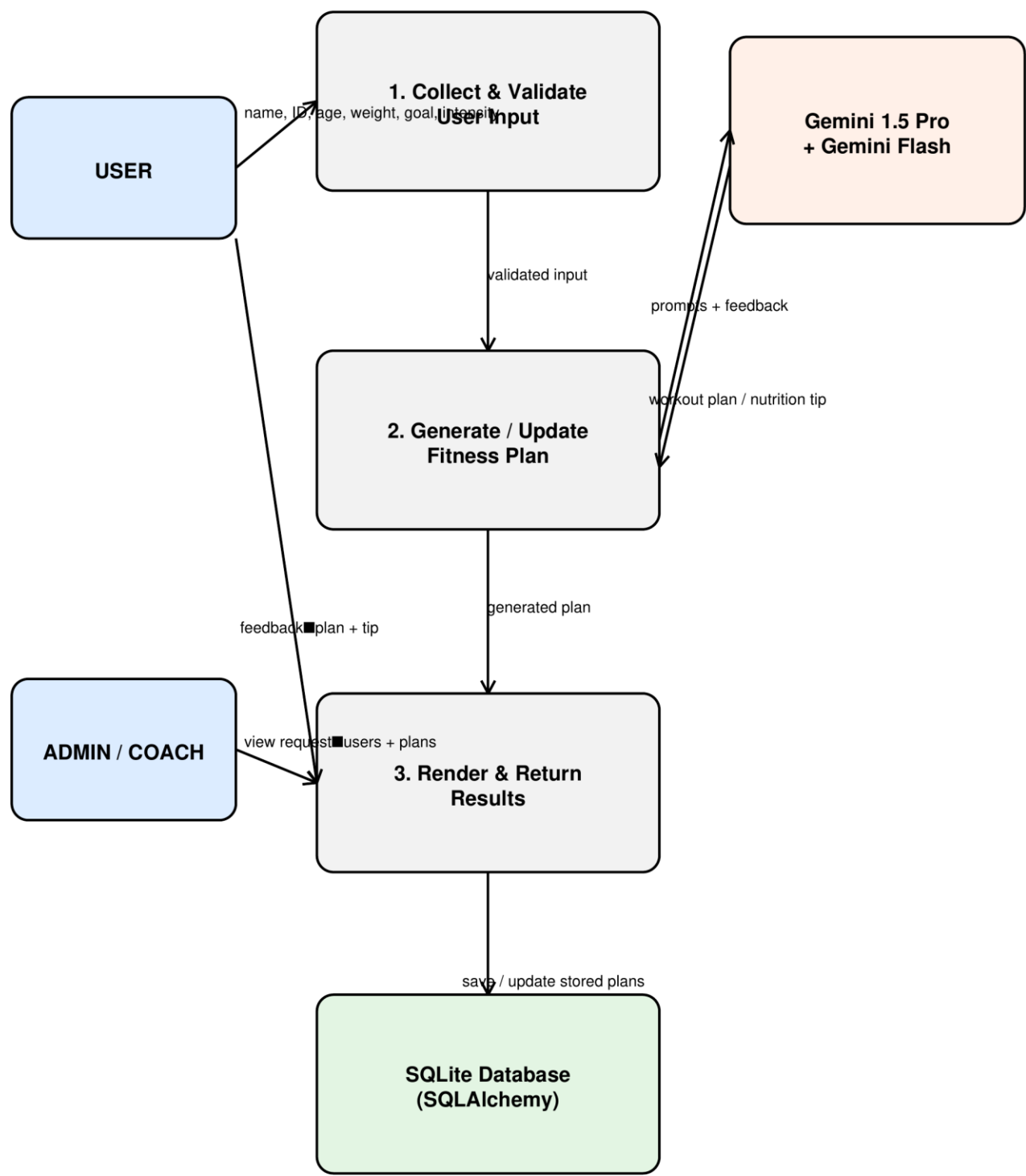
DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data.
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data.
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use.
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores.

Project-based DFD

The diagram on the following page represents the documented FitBuddy flow: user input is processed by FastAPI, workout and nutrition requests are sent to Gemini services, and user and plan records are stored in SQLite through SQLAlchemy.

FitBuddy AI – Data Flow Diagram



Flow Summary

1. The user enters name, user ID, age, weight, fitness goal and workout intensity. 2. FastAPI validates and processes the input. 3. Gemini 1.5 Pro generates the structured 7-day workout plan and updates plans from